MSC DATA SCIENCE DISSERTATION

Bottom-Up Multimodal Architecture for Hallucination-Suppressed Retail Captioning

Junyong Park

Supervisor: Dr Rita Borgo

BACKGROUND

The Edge VLM Constraint in Retail

Automated Shelf Monitoring

- Retailers lose significant revenue due to undetected out-of-stock items and compliance failures.
- Traditional Computer Vision requires costly, rigid retraining for every new product SKU.
- **Vision-Language Models (VLMs)** offer zero-shot recognition, adapting instantly without retraining.

The Edge Deployment Reality

- Cloud APIs (like GPT-4V or Gemini) are too slow, expensive, and raise GDPR privacy concerns for constant video feeds in physical stores.
- **Requirement:** We must use **Lightweight Local Models** (e.g., LLaVA-Phi3 3.8B) deployed on edge servers.

THE PROBLEM

Generative Hallucination

Physical Occlusion & Visual Ambiguity

- Retail environments are chaotic: harsh lighting, partially occluded products, and dense shelf layouts.
- When visual inputs are ambiguous, small edge VLMs rely heavily on pre-trained text priors rather than pixel evidence.
- **Consequence:** The VLM hallucinates items that are statistically likely to be there, but physically absent.



VLM GENERATED CAPTION:

> *"This display features various sandwiches including **Tuna, Chicken, and Egg Mayo**."*

- **Error:** The VLM hallucinates specific sandwich flavors based purely on language priors, ignoring the actual image.

ROOT CAUSE ANALYSIS

The Fallacy of Top-Down Injection

Why does Generative Hallucination occur?

Standard industry approaches attempt a **Top-Down** method: injecting specific retail taxonomies (e.g., GS1 SKUs) into the VLM prompt to restrict its answers.

Forced Taxonomy Mapping:

```
The model is forced to map a generic, blurry sandwich onto a
specific pre-defined SKU from the database (e.g., "Tuna" or "Egg
Mayo") to fulfill the taxonomy requirement.
```

The Resulting Error:

Because the visual evidence isn't clear enough, the VLM relies on its **Language Priors** to guess the closest match. It hallucinates specific brand names or flavors that simply aren't in the image, destroying caption integrity.

THE ARCHITECTURAL SOLUTION

Bottom-Up Physical Grounding (MOP)

The MOP Pipeline Principle

To fundamentally stop hallucination, we must restrict the VLM using **ONLY physical facts extracted directly from the image pixels** before text generation begins.

1. PHYSICAL ANCHORS (OCR)

We mechanically extract unforgeable text from the image, preventing the VLM from inventing product names.

2. CLIP GATEKEEPER

A mathematical filter that calculates Cosine Similarity, ruthlessly discarding any false candidate tags.

3. GENERATION RESTRAINT

An Open+Anchor prompting strategy forces the VLM to rely strictly on the validated tags without suppressing its descriptive fluency.

METHODOLOGY I

Hierarchical Tag Extraction (L1–L4)

The pipeline extracts visual evidence hierarchically (Macro → Micro) to cognitively constrain the VLM before prompt generation.



Case Study: sample_10000.jpg

L0 CONTEXT	Halloween Theme
L1 SCENE	Promotional Endcap
L2 FIXTURES	Display box
L3 PRODUCTS	Hard (0.75) Halloween Spooky Nuggets
L4 ATTRIBUTES	Promotion: limited offer at $2.49

Insight: Specific product names (L3) are successfully verified via physical evidence, eliminating the need f the VLM to guess.

METHODOLOGY II

K-Medoids Archetype Clustering (K=2)

Dynamic Prompt Routing via Physical Layout

Using Gower's Distance, the pipeline mathematically categorizes the shelf into objective physical archetypes to provide the true semantic context for accurate prompt routing.

Archetype 1 (e.g., Cluttered Display)

```
[Injected Prompt Template]
Focus heavily on promotional branding and complex product clusters.

VERIFIED PROMO ITEMS: {l3_tags}
EXPECTED FIXTURES: {l2_tags}
```

Archetype 2 (e.g., Standard Shelf)

```
[Injected Prompt Template]
Focus on linear shelf layout and uniform stocking.

VERIFIED STOCKED ITEMS: {l3_tags}
EXPECTED FIXTURES: {l2_tags}
```

This prevents "Logical Leaps" by adapting the VLM generation strategy dynamically to the environment.

VISUAL METHODOLOGY – STEP 1

Raw Image & OCR Extraction



Dense endcap with promotional items.

PaddleOCR Physical Extraction

We use PaddleOCR to mechanically extract all visible text blobs to establish unforgeable physical anchors.

```
Found Text Anchors:
- "Halloween"
- "Spooky Nuggets"
- "Limited Offer"
- "2.49"
```

This prevents the VLM from hallucinating arbitrary product names.

VISUAL METHODOLOGY - STEP 2

The CLIP Gatekeeper & Absence Filter

We calculate **Cosine Similarity** between the extracted text embeddings and visual crops. A strict threshold ($\tau = 0.25$) is applied to discard false anchors (Absence filter).

Valid Anchor Check:
`Sim(Image, "Spooky Nuggets") = 0.29`

✓ PASS (> 0.25 Threshold)

Hallucination Check (Injection):
`Sim(Image, "Diet Coke") = 0.18`

✗ REJECTED (< 0.25 Threshold)

By ruthlessly rejecting low-similarity tags, we mathematically block the injection of hallucinated product concepts before they reach the VLM.

VISUAL METHODOLOGY – STEP 3

MOP Prompt Injection (Open+Anchor)

Dynamic Template Routing

The VLM is commanded to describe freely (Open), but strictly adhere to the verified physical anchors.

```
[Injected Prompt]
Describe the scene in 2-3 natural sentences.
Detail the visual layout, colours, and themes.

VERIFIED PRODUCT ANCHORS (MUST MENTION):
- Halloween Spooky Nuggets
- 2.49

DO NOT CONFIRM (IF NOT VISUALLY EVIDENT):
- Beef Patties, Diet Coke.
```

Final VLM Caption

> *"The image shows a display featuring **Halloween Spooky Nuggets** in orange boxes... limited offer at **$2.49**."*

The VLM retains visual fluency (colors, layout) while adhering flawlessly to the verified physical facts.

EVALUATION DEMONSTRATION

Baseline vs. MOP Captioning



Sample 10000

Unconstrained Baseline

FAIL

> *"...Three boxes are on a shelf of M&Ms with a price tag... limited to the first 60 customers."*

- **Hallucination:** Invents "M&Ms" brand and fabricates a "first 60 customers" clause.

MOP Pipeline

PASS

> *"...display at the end of an aisle featuring Halloween Spooky Nuggets in orange boxes... limited offer at $2.49."*

- **Grounded:** Accurately anchors on OCR text without inventing surrounding elements.

EMPIRICAL VALIDATION

Quantitative Results (10,686 Images)

−30.1%

L-CHAIR_i Drop

Hallucination rate reduced from 70.47% to 40.36%, representing a **42.7% relative reduction** in false objects.

50.0%

Human Annotation (N=100)

MOP Win Rate: MOP won 50.0% vs Baseline 37.0% (13% ties). The low inter-rater agreement (κ = 0.036) highlights the complex human trade-off between the Baseline's descriptive fluency and MOP's strict factual safety.

71.1%

LLM Judge Win Rate

Gemini 3.5 Flash Judge closely mirrored human consensus.

[Multi-dim Rubric: Visual Accuracy, Factual Completeness, Safe Uncertainty]

DISCUSSION

Limitations & Future Work

Limitations

- **Cascading OCR Errors:** Because the VLM is anchored to tags, upstream garbled OCR text is forcibly integrated into the caption, causing structural errors.
- **Tag Starvation:** Severe lighting or blur causing OCR failure leads to no tags being passed, forcing the VLM to produce overly brief captions.

Future Work

- **Edge Hardware Acceleration:** As edge environments improve in capacity and latency, larger VLMs can be used to naturally offset OCR noise.
- **Multi-Agent Automation:** Upgrading to a multi-agent system where Vision and Analysis agents collaborate for human-level decision-making.

CONCLUSION

Evidence-First VLM Paradigm

This dissertation addresses the critical barrier to deploying Vision-Language Models in retail automation: **generative hallucination**. By abandoning conventional top-down taxonomy injection, we developed a fine-tuning-free Bottom-Up Multimodal Architecture that structurally blocks hallucination at its source.

Through the integration of OCR physical anchors and Explicit Absence Encoding, the system establishes a new **bottom-up epistemological framework**. It proves that by strictly prioritising verified physical evidence over generative priors, small edge VLMs can achieve highly reliable visual reasoning without the massive costs of parameter scaling.

THANK YOU

Junyong Park | King's College London